Supplementary Information

Equation discovery fails long before rate constants become unidentifiable in simulated adsorption columns

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Supplementary Table 1: where real columns sit on the Da map

The Discussion places real columns on the map with $\mathrm{Da} = k\,t_{\mathrm{stoich}}$, computed from inputs printed in four published studies. Where a study gives no stoichiometric time, $t_{\mathrm{stoich}} = Wq/\dot{n}$ is computed from the bed mass W, the uptake q at the feed and the molar feed of the adsorbate $\dot{n}$; where it does not say whether its volumetric flow is at column conditions or at 273.15 K and 1 atm, both are used and the range is given. "Rate constant" means different things in these studies: fitted to the breakthrough curves, computed from a macropore-diffusion correlation, or measured independently by frequency response and converted with the factor 15 of Glueckauf. The values are therefore order-of-magnitude estimates.

Supplementary Table 1. Damköhler number of published laboratory columns.

Study	System and conditions	k and t_{stoich}	Da
Lassitter et al. 2024, https://doi.org/10.1016/j.ces.2023.119430	water on MOF-303 pellets, packed bed, air at about 32 % relative humidity	k from frequency response (Fig. 4, read off), from the minimum at the isotherm step to the feed humidity; t_{stoich} from the first moment of the effluent curve (Fig. 10, read off)	96 to 3,100
Dantas et al. 2011, https://doi.org/10.1590/S0104-66322011000300018	CO_2 on zeolite 13X, two packed beds, 12 runs at 301 to 423 K	k computed from a macropore correlation (Table 7); t_{stoich} from W, q and flow (Tables 1, 2, 5)	200 to 23,000
Krishnamurthy et al. 2025, https://doi.org/10.1021/acs.iecr.5c00734	CO_2 (15 %) on MIL-91(Ti) pellets, two carrier gases	k fitted to breakthrough (Table 6); t_{stoich} from W, q and flow	280 to 560
Stampi-Bombelli et al. 2024, https://doi.org/10.1021/acs.iecr.4c01392	CO_2 (400 ppm and 5.6 %) on an amine sorbent, packed bed for direct air capture	k_1 fitted to breakthrough (Table 8); t_{50} as reported (Table 6)	7.6 to 9.7

Two further sources were read but not used for a number. For its coated tube, Lassitter et al. give no rate constant or stoichiometric time, but state that their group $k\rho n^{*}L/(vC_0)$, which is Da, was always very large in the cases they examined (Sec. 3.6). Büchner and Pini (2026, doi:10.1021/acs.langmuir.6c00250) imaged a 13X bed by X-ray tomography; their rate constant is computed from a correlation, and the form that sets its basis is given only in their supporting information, so it was not converted.

Supplementary Table 2: column, flow and physical parameters

Both adsorbents were solved in the same column geometry, particle size and superficial velocity. The step case also changes the bed porosity, particle density, heat of adsorption, feed and temperature, as recorded in erratum (b) of amendment A1. All values below are read from the code that built each solve, and the per-cell values from the ground-truth manifest.

The isothermal control uses the Langmuir column at Pe $\times$ 1 with no heat of adsorption. The feed-temperature van't Hoff factor is folded into b_{C0} and b_{H0} first, so the isotherm at T_{in} is the same as in the Langmuir cells and the bed stays at T_{in}.

Supplementary Note 1: exploratory analyses E2 to E10

Every analysis here was declared in writing before it ran (`EXPLORATORY_P2.md`) and enters no confirmatory test. E2 and E3 were declared before the verdicts; E4 to E7 after the corrected verdicts, prompted by a review of the draft; E8 after E6's corrected results; E9 after a second review; E10 after E9. E4 to E10 are post hoc and are labelled so wherever they are used. E1 (M1 at full scale) needs a graphics processor and has not run.

E2: the Kolmogorov–Arnold network on its declared subgrid

M8 ran in exactly the configuration declared before the freeze (raw variables c, q, T, q^*; sparsity weight 10^{-3}; default training steps; pykan 0.2.8) on all nine Da levels, $\sigma \in \{0.5, 2\}$ %, $\varepsilon \in \{0, 2\}$ %, both isotherms, Pe $\approx$ 94, O1, three replicates: 216 fits. It recovered the law in none of them, at every Da (0 of 24 at each level) and in every condition (0 of 27 in each). In 62 fits the extracted law contained both q and q^*, always with additional terms, most often a constant.

Isotherm	σ, ε	fits	law recovered	q and q^* selected
Langmuir	0.5 %, 0	27	0	9
Langmuir	0.5 %, 2 %	27	0	8
Langmuir	2 %, 0	27	0	4
Langmuir	2 %, 2 %	27	0	3
step	0.5 %, 0	27	0	13
step	0.5 %, 2 %	27	0	12
step	2 %, 0	27	0	7
step	2 %, 2 %	27	0	6

One fit stopped with an error inside the library (a failed least-squares step in its spline refit, `kan/spline.py`, `curve2coef`); it is counted as a failure.

E3: precision against coverage of the identifiability criterion

The pre-registered criterion calls k practically identifiable when the 95 % profile interval lies inside $[k/2, 2k]$. It measures precision and accuracy within a factor of two, not whether the interval contains the true value. On O1, 39 973 of 51 840 rows met the criterion, but only 5 042 intervals (10 %) contained the true k; 87 % of the rows called identifiable had an interval that missed it. The same held on O2 (coverage 9 %; 157 of 216 rows identifiable) and, less strongly, on O3 (coverage 23 %; 150 of 216). The intervals are narrow because the likelihood treats the measured isotherm and the observed c and T that enter q^*_{meas} as exact, while the estimate carries small biases from them and from the time resolution.

Supplementary Table 2. Parameters of the two simulated columns. "Langmuir" is the default physics; "step" is water on a MOF-303-like framework with a Type V isotherm.

Quantity	Langmuir	Step	Unit
Geometry and flow			
Column length L	0.10	0.10	m
Inner diameter D_{in}	0.05	0.05	m
Particle diameter d_p	0.002	0.002	m
Superficial velocity v	0.10	0.10	$\mathrm{m\,s^{-1}}$
Interstitial velocity $u = v/\varepsilon_t$	0.25	0.2857	$\mathrm{m\,s^{-1}}$
Residence time L/u	0.40	0.35	s
Bed and solid			
Total porosity ε_t	0.40	0.35	–
Particle density ρ_p	1200	1100	$\mathrm{kg\,m^{-3}}$
Solid heat capacity C_{ps}	1000	1000	$\mathrm{J\,kg^{-1}\,K^{-1}}$
Axial thermal conductivity k_z	0.10	0.10	$\mathrm{W\,m^{-1}\,K^{-1}}$
Gas and thermal boundary			
Gas density ρ_g	1.2	1.2	$\mathrm{kg\,m^{-3}}$
Gas heat capacity C_{pg}	1000	1000	$\mathrm{J\,kg^{-1}\,K^{-1}}$
Wall heat transfer coefficient h_w	10	10	$\mathrm{W\,m^{-2}\,K^{-1}}$
Feed and wall temperature $T_{\mathrm{in}} = T_w$	298.0	303.15	K
Equilibrium			
Saturation capacity $q_{\max}$	20.0	25.0	$\mathrm{mol\,kg^{-1}}$
Heat of adsorption ΔH	-50.0	-50.5	$\mathrm{kJ\,mol^{-1}}$
Cooperativity exponent n	1	4.0	–
Henry-site fraction f_H	1	0.06	–
Cluster affinity b_{C0}	1.720×10^{-9}	7.422×10^{-9}	$\mathrm{m^3\,mol^{-1}}$
Primary-site affinity b_{H0}	1.720×10^{-9}	3.711×10^{-10}	$\mathrm{m^3\,mol^{-1}}$
b_C at T_{in}	1.000	3.735	$\mathrm{m^3\,mol^{-1}}$
b_H at T_{in}	1.000	0.1868	$\mathrm{m^3\,mol^{-1}}$
Step position (cluster half saturation)	–	$P/P_0 = 0.15$ at 298.15 K	–
Feed concentration c_{in}	1.0	0.5043 (30 % RH)	$\mathrm{mol\,m^{-3}}$
$q^*(c_{\mathrm{in}}, T_{\mathrm{in}})$	10.00	21.90	$\mathrm{mol\,kg^{-1}}$
$q^*(c_{\mathrm{in}}, T_{\mathrm{in}})/q_{\max}$	0.500	0.876	–
Step midpoint loading θ (L2 only)	–	0.3552	–
Kinetics, dispersion and design			
Stoichiometric time t_{stoich}	7200.4	31 050.4	s
$\mathrm{Da} = k\,t_{\mathrm{stoich}}$, 9 levels	10^{-1} to 10^{3}, half-decade spacing		–
LDF coefficient k	1.389×10^{-5} to 0.1389	3.221×10^{-6} to 0.03221	$\mathrm{s^{-1}}$
Residence-time form kL/u	5.56×10^{-6} to 0.0556	1.13×10^{-6} to 0.0113	–
Default axial dispersion D_L (note)	2.675×10^{-4}	3.032×10^{-4}	$\mathrm{m^2\,s^{-1}}$
D_L multiplier ("Pe" level)	$\times 0.2$, $\times 1$, $\times 5$		–
D_L at the three levels	5.35×10^{-5}	6.06×10^{-5}	$\mathrm{m^2\,s^{-1}}$
	2.675×10^{-4}	3.032×10^{-4}	
	1.338×10^{-3}	1.516×10^{-3}	
$\mathrm{Pe} = uL/D_L$ at the three levels	467, 93.5, 18.7	471, 94.2, 18.8	–
L2 rate drop and width	–	0.8, 0.05	–
Solves			
Grid cells N_z, stored times	400, 400	400, 400	–
Horizon (until outlet $c/c_{\mathrm{in}} > 0.95$)	4 or 8 t_{stoich}	4 or 8 t_{stoich}	–
Time integrator	BDF, rtol 10^{-6}, atol 10^{-10}, upwind advection		–
Cells (L1, L2, isothermal control)	27, 0, 9	27, 27, 0	–

Note: the default axial dispersion is $D_L = 0.7 D_m + 0.5\,d_p u$ with $D_m = 2.5 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$.

Da	identifiable (of 5 760)	interval contains k	identifiable but misses k
0.1	5 760	425	92.6 %
0.316	5 760	454	92.1 %
1	5 760	645	88.8 %
3.16	5 760	1 042	81.9 %
10	5 760	1 509	73.8 %
31.6	5 564	661	88.1 %
100	3 318	226	93.2 %
316	1 843	47	97.4 %
1000	448	33	94.0 %

Profiles whose likelihood peaked on the upper edge of the k grid (807 rows, all at Da ≥ 100) raise an error, count as not identifiable and cover nothing.

E4: matched success criteria (post hoc)

Discovery must return exactly the terms q and q^*_{meas} with coefficients in the ratio -1 within 10 %, while identifiability asks only for a 95 % interval inside $[k/2, 2k]$. To see whether this asymmetry makes the gap, H2b was recomputed with the pre-registered analysis unchanged except for the success of each row: (a) discovery relaxed to exactly the terms $\{q, q^*_{\mathrm{meas}}\}$ with a negative coefficient on q (the ratio test dropped; the q^*_{meas} coefficient is not stored, so its sign could not be checked); (b) identifiability strict, interval inside $[0.9k, 1.1k]$; (c) identifiability by the point estimate, the interval's geometric midpoint inside $[k/2, 2k]$, which does not rely on the interval's calibration (E3 above); (d) (a) and (c) together. Boundaries are point estimates by the same logistic estimator.

	Success rule	slices with a verdict	gap (decades of Da)
pre-registered	as frozen	37	1.60 to 3.72
a	discovery relaxed	40	1.60 to 3.72
b	interval inside $[0.9k, 1.1k]$	37	0.89 to 2.94
c	point estimate inside $[k/2, 2k]$	37	1.79 to 3.75
d	a and c	40	1.79 to 3.75

Every slice with a verdict, under every rule, read *discovery fails before identifiability*, and both positive controls passed. Relaxing discovery added three slices, all with 5 % noise and 2 % isotherm error, and moved no existing gap by more than 0.16 decade; the strict interval shortened every gap, by 0.52 to 1.01 decade; the point-estimate rule lengthened them by at most 0.27 decade. The gap is therefore not an artefact of the two success rules.

E5: conditioning of the library against Da (post hoc)

On the O1 observations without noise or isotherm error, the columns of the library were standardised and the singular values of the pair $\{q, q^*_{\mathrm{meas}}\}$, the two terms of the true law, computed for every cell. The smaller singular value relative to the larger fell steadily with Da, by about a factor of three per half decade: from 0.73 (Langmuir) and 0.78 to 0.79 (step) at Da = 0.1, through 0.21 to 0.25 at Da = 1 and 0.051 to 0.060 at Da = 10, to 0.0007 to 0.0013 at Da = 1000, nearly the same for the three Péclet numbers. The condition number of the whole library was between about 230 and 5 700 and was not monotone in Da: it was smallest near Da = 1 to 10 and largest at Da = 1000. As Da grows the uptake follows the measured isotherm ever more closely, so the direction that distinguishes q from q^*_{meas}, the one the law needs, carries ever less of the signal.

E6: choosing among three candidate laws (post hoc)

Three one-parameter laws were fitted to each O1 observation by the output-error likelihood of the identifiability profile: the LDF law $k(q^*_{\mathrm{meas}} - q)$, the quadratic driving force $k({q^*_{\mathrm{meas}}}^2 - q^2)/(2q)$ and the pseudo-second-order law $k(q^*_{\mathrm{meas}} - q)^2$. Each is integrated by implicit Euler with four substeps per interval along the same upsampled $q^*_{\mathrm{meas}}(t)$ path as the identifiability profile; its k is profiled on `logspace(-7, 1, 81)` and refined by a bounded one-dimensional search between the grid neighbours of the grid optimum. The declared procedure read k off the grid and took one implicit step per upsampled interval; the refinement and the substeps were added after the first rows (B91, Supplementary Note 2), and every row was recomputed. With one parameter each, the Bayesian information criterion chooses the largest likelihood. The subgrid was Pe ≈ 94, both isotherms, all nine Da, $\sigma \in \{0.5, 2\}$ %, $\varepsilon \in \{0, 2\}$ %, estimation replicates 15 to 59 (3 240 fits of each law). The table gives the fraction of replicates in which the LDF law was chosen. Wherever it was chosen at Da ≤ 31.6, its k lay within $[k/2, 2k]$; at Da ≥ 100 this held in 22 of the 32 replicates that chose it.

		Da								
Isotherm	σ, ε	0.1	0.32	1	3.2	10	32	100	316	1000
Langmuir	0.5 %, 0	1.00	1.00	1.00	1.00	1.00	0.69	0	0	0
Langmuir	0.5 %, 2 %	1.00	1.00	0.98	0.96	0.80	0.47	0.51	0.04	0.09
Langmuir	2 %, 0	1.00	1.00	1.00	1.00	1.00	0	0	0	0
Langmuir	2 %, 2 %	1.00	1.00	0.98	0.96	0.82	0.04	0	0	0
step	0.5 %, 0	1.00	1.00	1.00	1.00	1.00	0	0	0	0
step	0.5 %, 2 %	1.00	1.00	1.00	1.00	0.78	0.31	0.02	0.04	0
step	2 %, 0	1.00	1.00	1.00	1.00	1.00	0	0	0	0
step	2 %, 2 %	1.00	1.00	1.00	1.00	0.78	0.16	0	0	0

With an exact isotherm the choice was nearly unanimous at every level: the LDF law up to Da = 10, the quadratic driving force at Da = 31.6 (except 31 of 45 replicates for the Langmuir sorbent at 0.5 % noise, which chose the LDF law), the quadratic driving force or the pseudo-second-order law at 100 (for the Langmuir sorbent at 2 % noise, 41 replicates the pseudo-second-order law and 4 the quadratic driving force), and the pseudo-second-order law at 316 and 1000. Except for the Langmuir sorbent at Da = 31.6, the choice did not depend on the noise level; E9 tests whether noise in the forcing $q^*_{\mathrm{meas}}(t)$ is nevertheless the cause, and E8 whether the sampling rate is. With 2 % isotherm error the LDF law was chosen in some replicates where the exact isotherm gave none, most often for the Langmuir sorbent at 0.5 % noise and Da = 100. Since E9 shows that the exact-isotherm choice there is set by a systematic misfit, not by noise, we read these choices as one systematic error offsetting part of another, not as evidence for the law.

E7: a shifted isotherm step (post hoc)

The pre-registered isotherm error is a smooth random multiplicative ripple. A real water isotherm can also be wrong in where its step lies, an error of a different shape. For the step isotherm (O1, Pe ≈ 94, all nine Da, $\sigma \in \{0.5, 2\}$ %, replicates 0 to 59), the measured isotherm was given a step shifted by ± 1 and ± 2 % relative humidity, with the fields unchanged; discovery (the six sparse solvers in both forms, best method chosen on replicates 0 to 14) and O1 identifiability were computed as on the grid. The identifiability rows were recomputed after correction B90.

σ	step shift (% RH)	best discovery success, any Da	$\mathrm{Da}^*_{\mathrm{ident}}$	median $\hat{k}/k$ at Da = 10
0.5 %	−2	0.00	5.6	0.48
0.5 %	−1	0.00	57	0.67
0.5 %	+1	0.00	17	1.76
0.5 %	+2	0.00	5.6	4.98
2 %	−2	0.00	5.6	0.48
2 %	−1	0.31	27	0.67
2 %	+1	0.27	17	1.74
2 %	+2	0.00	5.6	4.31

Discovery reached 50 % at no rate under any shift. Identifiability held only to Da ≈ 17 to 57 with a 1 % shift and to Da ≈ 5.6 with a 2 % shift (all-to-none between Da = 3.16 and 10, so interval-censored), against Da ≈ 172 to 562 for the same noise levels with an exact isotherm (E8, 200 samples). The intervals stayed narrow (median upper over lower bound at most 1.16 per level, and at most 1.19 in any replicate, at Da ≤ 31.6), while the median estimate at a 1 % shift was off by 10 to 15 % at Da = 1 and 20 to 33 % at Da = 3.16, at both noise levels: precise, but wrong. The estimate was biased, downwards when the step was placed too low and upwards when too high, by a factor that grew with Da. For positive shifts at high Da the likelihood peaked on the upper edge of the k grid ($k = 10\,\mathrm{s}^{-1}$); these 839 rows raise an error and count as not identifiable. A misplaced step therefore limits the estimate of k as well as discovery. The grid's smooth isotherm error moved $\mathrm{Da}^*_{\mathrm{ident}}$ by less than 0.2 decade at $\varepsilon = 0.5\,\%$ but by 0.43 to 0.92 decade at 5 % ($\sigma = 0.5\,\%$), so the shape of the error matters as well as its size.

E8: sampling rate (post hoc)

E6 chose the wrong law at Da ≥ 100 with an exact isotherm at both noise levels, and at Da = 31.6 in every case but the Langmuir sorbent at 0.5 % noise. E8 asks whether the time resolution is the cause. On O1 (Pe ≈ 94, both isotherms, all nine Da, $\sigma \in \{0.5, 2\}\,\%$, $\varepsilon = 0$, replicates 0 to 59), every quantity was computed at the grid's 200 observation times and at 400 (every stored time of the ground-truth fields, with the same noise per sample): discovery by the six sparse solvers in both forms, the best method chosen on replicates 0 to 14, O1 identifiability with the pre-registered criterion, and E6's choice among three laws on replicates 15 to 59. Every method ran unchanged, so windows defined in samples (the 21-sample derivative filter, the weak-form supports) span half the time at 400 samples. The 4 320 rows ran on Kaggle; four sampled rows recomputed locally agreed exactly on discovery and to a relative 10^{-6} on identifiability and model choice. No row had an error. The table gives the 50 % crossings (Da^*) and the gap $\log_{10}(\mathrm{Da}^*_{\mathrm{ident}}/\mathrm{Da}^*_{\mathrm{disc}})$.

Isotherm	σ	samples	best method	$\mathrm{Da}^*_{\mathrm{disc}}$	$\mathrm{Da}^*_{\mathrm{choice}}$	$\mathrm{Da}^*_{\mathrm{ident}}$	gap
Langmuir	0.5 %	200	M4, weak	7.24	35	356	1.69
Langmuir	0.5 %	400	M2, weak	32	57	562	1.25
Langmuir	2 %	200	M4, weak	1.05	17	74	1.84
Langmuir	2 %	400	M4, weak	4.93	30	141	1.46
step	0.5 %	200	M4, weak	4.18	17	562	2.13
step	0.5 %	400	M4, weak	10	57	–	–
step	2 %	200	M4, weak	0.85	17	172	2.30
step	2 %	400	M4, weak	3.02	36	172	1.75

With 400 samples every boundary moved to higher Da or stayed: discovery by 0.40 to 0.67 decade, the choice among laws by 0.21 to 0.52 decade, identifiability by 0 to 0.28 decade (for the step isotherm at 0.5 % noise, beyond the grid). Discovery still failed first wherever both boundaries

existed, by at least 1.25 decades. The LDF law was chosen in every replicate up to Da = 31.6 at 0.5 % noise with 400 samples, but in none at Da = 100 at either rate, so the time resolution explains the failure of the choice at Da = 31.6 but not above it.

E9: noise in the forcing of the choice among laws (post hoc)

E6 drives each law with $q^*_{\mathrm{meas}}(t)$ computed from the noisy observed c and T, so its likelihood carries errors in the variables. E9 repeated E6 unchanged on its exact-isotherm rows at Da = 31.6 to 1000 (Pe $\approx$ 94, both isotherms, $\sigma \in \{0.5, 2\}$ %, estimation replicates 15 to 59; 720 fits of each law), with $q^*_{\mathrm{meas}}(t)$ computed from the noise-free c and T at the same probes and times and q observed with noise as before. One row was run to time the check before it was declared; it is recomputed here. The reading was fixed in advance: where the fraction of replicates choosing the LDF law rises to at least two thirds, the failure is attributed to noise in the forcing; where it stays at or below one third, it is not. No row had an error. The table gives the fraction with noise-free forcing and, in brackets, E6's.

		Da			
Isotherm	σ	31.6	100	316	1000
Langmuir	0.5 %	1.00 (0.69)	0 (0)	0 (0)	0 (0)
Langmuir	2 %	0.69 (0)	0.11 (0)	0.22 (0)	0.20 (0)
step	0.5 %	0 (0)	0 (0)	0 (0)	0 (0)
step	2 %	0 (0)	0 (0)	0 (0)	0 (0)

Noise in the forcing explains the failure only for the Langmuir sorbent at Da = 31.6, where twice the sampling rate also restored the choice at 0.5 % noise (E8). For the step isotherm at every level, and for both isotherms from Da = 100 on, the wrong law was still chosen. A diagnostic made after E9 and not declared (`p2_diag_e9_noisefree.py`, replicate 15) removed the noise from q as well: the LDF law was still not chosen at Da = 100 or 1000, although its fitted k lay within 2 % of the true value at Da = 100 for both isotherms. The failure from Da = 100 on is therefore a deterministic misfit between the integrated laws and the sampled data, not an effect of noise; E10 asks whether finer sampling removes it.

E10: time resolution behind the misfit (post hoc)

At the grid's 200 observation times kh is about 2 at Da = 100 and 20 at Da = 1000, so q lags q^*_{meas} by less than one sampling interval, and every law is driven by $q^*_{\mathrm{meas}}(t)$ reconstructed between samples. Four cells (Pe $\approx$ 94; both isotherms; Da = 100 and 1000) were solved again with the same solver and settings but 3 991 stored times, every tenth of which is one of the grid's 400; at those times the fields agreed exactly with the stored fields. E6's choice (unchanged, as in the diagnostic above) was then made on noise-free data at the 20 probes at increasing time resolution. The reading was fixed in advance: where the LDF law is chosen at 3 991 times but not at 200, the misfit is attributed to the reconstruction of q^*_{meas} between samples. The table gives the chosen law and, in brackets, the LDF law's fitted k over the true value.

Isotherm	Da	200 (grid)	400 (grid)	1 996	3 991
Langmuir	100	QDF (0.98)	LDF (1.05)	LDF (1.01)	LDF (1.01)
step	100	QDF (1.01)	QDF (1.08)	LDF (1.01)	LDF (1.01)
Langmuir	1000	PSO (0.51)	LDF (0.96)	LDF (1.02)	LDF (1.04)
step	1000	QDF (0.45)	QDF (1.12)	QDF (1.08)	QDF (1.04)

At Da = 100 the LDF law won once sampling was fine enough, from 400 times for the Langmuir sorbent and from 1 996 for the step isotherm; for the Langmuir sorbent at Da = 1000 it won from 400 times. In these three cells the misfit is a matter of sampling. For the step isotherm at Da = 1000 the quadratic driving force was still chosen at 3 991 times, where kh is still about 1; there the cause is unresolved. The declared design spaced 200 and 400 times evenly on the finer grid, which shifts them by up to half a grid interval from the grid's own times, so the grid's exact times were added after the first rows. The evenly spaced rows agree with the grid's except for the Langmuir sorbent at Da = 1000 and 200 times, where the evenly spaced times chose the LDF law, 38 log-likelihood units ahead of the quadratic driving force, while the grid's own times chose the pseudo-second-order law: at that resolution the choice turns on where the samples fall. These are noise-free data; with noise, twice the sampling rate did not restore the choice at Da = 100 (E8).

Supplementary Note 2: corrections made during the study

Every defect found in the design, code or text after it was written is recorded in a public ledger (`RETRACTIONS_P2.md`) with its cause, its fix and how it was found. Entries B76 and B79 to B84 were found before any grid result existed and reached no result; they corrected the analysis code, one method's inclusion gate, cost statements, citations and descriptions of the ground truth. B85, a provenance field missing from three results files, was found while the grid ran and affects no result. Six entries were found while the grid ran or after its first verdicts and changed what was computed (B86 to B91): three changed confirmatory identifiability rows (B87, B89, B90), one capped the O2 and O3 solves without changing a finished row (B86), one the M1 demonstration (B88) and one an exploratory analysis (B91). Each is described here with what it changed.

A5: a cap on each column solve (B86)

On O2 and O3, every point of the identifiability profile is a full column solve. With an isotherm error $\varepsilon > 0$, the measured isotherm evaluates its error term at c/c_{in} clipped to $[0, 1]$, and at large trial k the stiff solver kept crossing that kink: one O3 row made no progress for six days, and two O2 rows of the same cell took 39 and 42 hours. Amendment A5, made while the grid ran, capped each solve at 600 s and scored a capped solve as a failed solve. Every capped k is recorded with whether it lies next to the reported interval. In the finished O3 run, two rows had capped solves (six trial values of k each, all far above the interval, none adjacent to it). No reported interval depends on a capped solve.

A6: the range of the O1 profile (B87)

The pre-registered positive control requires discovery and identifiability to succeed at the lowest Da with no noise and no isotherm error; an isotherm that fails it is withheld from every verdict. As first run, the step isotherm failed: discovery succeeded in 135 of 135 estimation replicates and identifiability in none. The cause was the O1 profile's search range, 10^{-5} to $10\,\mathrm{s}^{-1}$, taken from a statement that holds for the Langmuir column only. The step cells at Da = 0.1 have $k = 3.2 \times 10^{-6}\,\mathrm{s}^{-1}$, below the range, so their likelihood peak was cut at the range's edge and every interval sat at $3.1\,k$. The stored fields obey the LDF law exactly there, and with a range that contains k the interval is $[1.0034, 1.0038]\,k$. Amendment A6, made after the first verdicts were seen, widened the range to 10^{-7} to $10\,\mathrm{s}^{-1}$ at the same density, made a peak on either edge an error, and recomputed all 51 840 O1 identifiability rows; the analysis was then rerun unchanged. A6 changes

the range of a numerical search, chosen to contain every k of the design, and no criterion, threshold, estimand or verdict word.

Refinement points of the O1 profile (B89)

While drafting Supplementary Note 3, the O1 profile was found to refine with 21 points per pass, where the pre-registration declares 41 for this profile (21 is declared for the O2 and O3 profiles). Because the reported ends are the outside neighbours of the last points inside the interval, fewer points can widen an interval by up to one fine step, about 10 % instead of about 5 % for an interval of 0.6 decade. In the first run, 1 464 of the 13 849 non-identifiable O1 rows lay within 15 % of the $[k/2, 2k]$ limits with the other end inside. The A6 recomputation was 12 of 54 cells in when this was found; it was stopped, the refinement was set to 41, and the recomputation was restarted from the beginning, so every corrected O1 row uses the declared settings.

The exact update of the O1 likelihood (B90)

The O1 likelihood integrates the LDF law $\mathrm{d}q/\mathrm{d}t = k(q^* - q)$ over each interval h between upsampled observations, with q^* linear in time across the interval. The exact update is

$$q_1 = e\, q_0 + (1 - e)\, q_1^* - (q_1^* - q_0^*)\,(g - e), \qquad e = \mathrm{e}^{-kh}, \quad g = (1 - e)/(kh).$$

The code subtracted $(q_1^* - q_0^*)(1 - g)$ where the exact solution needs $(q_1^* - q_0^*)(g - e)$. Both coefficients equal $kh/2$ to first order and differ by $(kh)^2/6$ at the next, so the original comparison with an ODE-solver likelihood, made at small kh, passed. At $kh = 1$ the error of a single update was 0.22, and at $kh = 10$ it was 1.68, for a worst case in which q^* changes by 2.1 within one interval; as kh grows the update lags q^* by one step. The likelihood therefore misfit the data at high Da: with exact data the estimate at Da = 316 was 1.1 to 24 times k, which an earlier draft had attributed to the time resolution of the data. The error was found in a later review of the code, derived, and confirmed against an ODE solver. The update was corrected and is now tested against `solve_ivp` for kh from 10^{-4} to 100. Because the pre-registration states that the O1 likelihood is integrated exactly, the correction brings the code into line with it; amendment A7 records it. All 51 840 O1 identifiability rows were recomputed with the corrected code (on cloud CPU sessions; 24 rows drawn at random were recomputed locally and agreed to a relative 10^{-6}), and the analysis was rerun unchanged. 916 rows became identifiable and 1 834 non-identifiable, all at Da $\geq$ 31.6. With exact data the estimate is now within 5 % of k up to Da = 100 and 0.39 to 0.91 times k at Da = 316 and 1000; that remaining bias is the time resolution. Of the 71 identifiability boundaries present in both runs, 36 moved to lower Da and none to higher; 22 more came onto the grid. Seven more slices gained both boundaries, and every verdict word was unchanged (table below).

The rate constants of the model-selection check (B91)

The exploratory model-selection check (E6) first took each candidate law's rate constant from a grid of 10 points per decade without refinement. At 0.5 % noise a k 10 % from a law's optimum costs about 10^5 log-likelihood units, far more than the differences between laws, so the chosen law followed which law had a grid point near its optimum: for the Langmuir sorbent at Da = 0.1 with an exact isotherm, the pseudo-second-order law was chosen in all 45 replicates although the data were generated by the LDF law. Each law's k is now refined by a bounded one-dimensional search between the grid neighbours of the grid optimum, and each law is integrated with four implicit-Euler substeps per interval. E6 was rerun; its first rows are kept.

M1 kept its best parameters (B88)

Amendment A4 says the M1 demonstration keeps the best parameters seen during training. In four of its nine fits (Da = 31.6, 100, 316 and 1000), a solve of the differentiable column exhausted its step budget after one to two hours of training; only the time budget was handled, so the error ended the fit with no readout, and it was recorded as a failure. The training loop now stops on such a failure and keeps the best parameters seen; a failure on the first evaluation is still recorded as an error. The four fits were rerun with this code. Each stopped on a solver failure after 9 evaluations, with a training loss near 0.1, about a hundred times the loss the fits at Da $\leq$ 3.16 reached, and none recovered the law. M1 therefore succeeded in one of its nine fits (Da = 3.16, symbolic readout) in both runs. M1 enters no confirmatory test, so no verdict depends on it.

As-run, intermediate and corrected verdicts

All three sets of verdicts are public (`results/p2_verdicts_asrun.json`, `results/p2_verdicts_preB90.json` and `results/p2_verdicts.json`). As first run, the Langmuir positive control passed (discovery and identifiability 135 of 135) and the step control failed, so the step isotherm was withheld from all 48 of its H2b slices. Among the 48 Langmuir slices, 20 had a verdict, and in all 20 discovery failed before identifiability; 28 had no boundary on the grid for discovery or identifiability. H2a and H2c had no verdict. The intermediate verdicts, after A6 and B89 but before B90, are those an earlier draft reported. The corrected verdicts are the primary result, because only they compute the pre-registered estimand, the 95 % profile interval of the exactly integrated likelihood, for every cell.

	As first run	After A6, B89	Corrected (primary)
Positive control, Langmuir	passed	passed	passed
Positive control, step	failed (0/135 identifiable)	passed	passed
O1 rows identifiable (of 51 840)	37 991	40 891	39 973
H2b slices with a verdict	20 (Langmuir only)	30 (20 + 10 step)	37 (23 + 14 step)
H2b overall	fails before	fails before	fails before
H2b gap, Langmuir (decade)	−1.60 to −3.43	−1.60 to −3.43	−1.60 to −3.36
H2b gap, step (decade)	withheld	−2.23 to −2.83	−2.05 to −3.72
H2a	no verdict	no verdict	no verdict
H2c	no verdict	no verdict	no verdict

Neither correction changed what was found. A6 and B89 changed which isotherm could be judged: the 2 900 O1 rows that became identifiable are step rows, nearly all in the cells A6 corrected; among the Langmuir rows, the 41-point refinement (B89) turned 24 rows identifiable and 17 non-identifiable, and the Langmuir H2b gaps moved by at most 0.11 decade (in one slice, where the identifiability boundary moved from Da $\approx$ 57 to 74), with every verdict unchanged. B90 changed only rows at Da $\geq$ 31.6 (above). Of the 71 identifiability boundaries present in both runs, it moved 36 to lower Da, by up to 0.8 decade where the isotherm error was largest, and none to higher; the gaps of the 30 slices judged before shrank by at most 0.25 decade. Seven slices that already had a discovery boundary gained an identifiability boundary, because k was no longer identifiable at the highest levels (three without noise and with 0.5 % isotherm error, three at 0.5 % noise with an exact isotherm, one at 0.5 % noise and isotherm error); these boundaries lie between Da = 316 and 1000, where the estimate is biased by the time resolution, and the check without those levels (main text) covers them. Every slice judged before kept its verdict, and every new slice has the same verdict.

Supplementary Note 3: methods, settings and known-answer tests

Deviations from the pre-registration

Where the code and the frozen pre-registration differ, the paper reports what the code did. The differences, found by checking the code against the text line by line before submission, are:

- *Weak-form test-function support.* The support is chosen per replicate and per solver (not per cell) by the held-out weak-system residual; when no held-out probe exists it falls back to 61 samples, and when no support is admissible to 31.
- *Selection floor in the weak form.* Exact best-subset selection floors the residual variance at 2×10^{-3} times the standard deviation of the regression target, which in the weak form is the weak target, not $\mathrm{d}q/\mathrm{d}t$.
- *Errors-in-variables weights.* M5 weights each column by the mean of its error standard deviation over the rows, and takes the channel ranges from the observed data.
- *M1 stopping rule.* Besides 50 iterations or 2 h, the optimiser also stops at 60 function evaluations; the fits at $\mathrm{Da} = 0.1$ and 0.316 stopped there.
- *Likelihood scale on O2 and O3.* The noise standard deviation in the O2 and O3 likelihoods uses the true channel range (the range the noise was drawn with); on O1 it uses the observed range.
- *Symbolic-regression gate.* The inclusion gate's noise is relative to the local value; the grid's noise is relative to the channel range.
- *Positive control.* The three Péclet numbers are pooled in each isotherm's easiest cells; the pre-registration names one easiest cell.
- *Order of reporting.* The pre-registration says O3 is reported first; the paper leads with O1, where the confirmatory test lives.
- *Analyses run late or not run.* The held-out fidelity of the isotherm-free library (Lib-A) was declared but computed only after the verdicts (below). Placing the closest prior study on the map was done only after the verdicts (Discussion; its Damköhler number from its published equations). Applying the best method to measured MOF-303 data (declared exploratory) was not done: no method discovered the law from outlet data alone in the simulations, so an outlet-only application could not be interpreted, and the uptake kinetics of the cited MOF-303 study exist only as figures. The inclusion probabilities of the ensemble's terms were not stored, and WENDy-IRLS and ODR-BINDy were run only on their authors' published examples (NOT RUN on the grid). The Kolmogorov–Arnold network was not run on the grid because its fit exceeded the declared cost gate on this machine, a machine limit.

Observation models

Every method and every profile likelihood read data only through one function. O1 gives c, q and T at 20 equally spaced probe positions. O2 gives c and T at the same positions, with q hidden. O3 gives the outlet $c(t)$ and $T(t)$ only. All three take 200 times, evenly spaced over the solved horizon. The probe and time indices are rounded from a linear spacing of the stored 400 cells and 400 times.

Noise σ and isotherm error ε

Noise perturbs the observed channels only. Each observed channel gets independent Gaussian noise with standard deviation σ times that channel's range over the whole ground-truth field of the cell, not over the observed points. The levels were $\sigma \in \{0, 0.005, 0.02, 0.05\}$.

The isotherm error perturbs only the equilibrium the method is given. It does not touch the data. The method receives $q^*_{\text{meas}}(c,T) = q^*(c,T)\,[1+\varepsilon\, g(c/c_{\text{in}})]$, where $g(x) = \sum_{j=1}^{4} a_j \sin(j\pi x + \phi_j)$ divided by its root-mean-square on 2001 points of $[0,1]$, with a_j standard normal and ϕ_j uniform on $[0, 2\pi)$. The argument c/c_{in} is clipped to $[0,1]$. The function g is drawn once per (cell, replicate) and does not depend on σ or on the observation model. The levels were $\varepsilon \in \{0, 0.005, 0.02, 0.05\}$.

Every draw is seeded by the first 16 hexadecimal digits of a SHA-256 hash of its keys. The noise keys are (cell, replicate, σ, observation model); the isotherm-error keys are (cell, replicate). So the same inputs always give the same observation. A self-test checks that $\sigma = 0$ returns the stored field exactly, $\varepsilon = 0$ returns the true isotherm, the noise standard deviation is within 5 % of σ times the range, the isotherm error has root-mean-square ε to 10^{-3}, draws repeat, O2 hides q and O3 is outlet only.

Each cell had 60 replicates for every method run at the full count. Replicates 0 to 14 choose the best method of a slice; replicates 15 to 59 estimate every boundary.

Known-answer system

Most method tests use one synthetic system that is never a grid cell: $\mathrm{d}q/\mathrm{d}t = k(q^*(c) - q)$ with $k = 0.02\,\mathrm{s}^{-1}$, a Langmuir $q^*(c) = 5 \cdot 3c/(1+3c)$, a logistic inlet ramp $c(t) = 1/(1+\exp(-(t - 0.4\,t_{\text{end}})/(0.05\,t_{\text{end}})))$, $t_{\text{end}} = 600\,\mathrm{s}$, constant $T = 298\,\mathrm{K}$, and 400 samples unless a test says otherwise. It is solved with rtol 10^{-10} and atol 10^{-12}.

Shared settings: library, derivative and selection floor

Lib-B is $\{1, c, q, T, cq, q^2, c^2, q^*_{\text{meas}}\}$, with q^*_{meas} evaluated at the observed c and T. Lib-A is Lib-B without q^*_{meas}. Under L1 the true law in Lib-B is the support $\{q^*_{\text{meas}}, q\}$. The strong-form target $\mathrm{d}q/\mathrm{d}t$ is a Savitzky–Golay derivative with window 21 and polynomial order 3 on the uniform time grid. Rows from all probes are stacked, so one law is fitted for the whole column.

Every sparse solver works on columns divided by their standard deviation (a constant column keeps scale 1). Coefficients are mapped back to the original units before scoring.

M2 and M2b: sparse regression in the weak and strong forms

The strong form (M2b) regresses the Savitzky–Golay derivative on Lib-B. The weak form (M2) multiplies the law by test functions $\phi(t) = ((t-a)(b-t))^8$ on windows of *support* samples, with a stride of support/4 (integer division), and integrates by parts with trapezoid weights, so the target needs no derivative of the data. Each ϕ is scaled to a maximum of 1. The support is swept over $\{31, 61, 121\}$ samples. For each support the system is fitted on the probes not held out and scored by $\|G\theta - b\|/\|b\|$ on the held-out probes, which are every fifth probe (indices 2, 7, 12 and 17 of 20). A support is admissible only if its training system has at least 3 equations per library term. The chosen support is then refitted on all probes. Both forms run all three solvers below.

Solvers. Sequentially thresholded least squares (STLSQ) uses ridge parameter 10^{-6} and drops a term when its standardised coefficient falls below 0.05 of the largest, for at most 20 iterations. The ensemble (M3) fits STLSQ with the same threshold on 100 bootstrap samples of 60 % of the rows,

drawn with replacement from a fixed seed 0, keeps a term when it appears in at least 60 % of the fits, and reports its median coefficient. Exact best-subset selection (M4) enumerates every support of 1 to 4 terms, fits each by least squares, and picks the smallest BIC $= n \ln \max(\mathrm{RSS}/n,\ (2 \times 10^{-3}\,\mathrm{sd}(y))^2) + m \ln n$, with m the number of terms. M4 has no sign constraints.

Known-answer tests. On the known-answer system at zero noise, M4 recovers L1 with $|\hat{k} - k|/k < 0.02$, and STLSQ and the ensemble both recover L1. The weak form with M4 recovers L1 at $\sigma = 0$ and at $\sigma = 2\,\%$ with $|\hat{k} - k|/k < 0.05$, and at $\sigma = 2\,\%$ its k error is smaller than the strong form's. On a five-probe, 200-sample version, every solver recovers L1 in both forms through the grid's own pipeline. The residual floor 2×10^{-3} was the smallest of $\{1, 2, 5\} \times 10^{-3}$ that passed at 200 samples; the derivative's error there is 8×10^{-4} of $\mathrm{sd}(\mathrm{d}q/\mathrm{d}t)$.

M1b: uptake from the gas balance (O2)

On O2, q is recovered from the gas-phase balance of the solver, $\partial q/\partial t = (D_L c_{zz} - u c_z - c_t)/S$ with $u = v/\varepsilon_t$ and $S = (1 - \varepsilon_t)\rho_p/\varepsilon_t$, then integrated in time from $q = 0$ by the trapezoid rule. c_t is the Savitzky–Golay derivative above; c_z and c_{zz} are centred finite differences over the probe positions. The material and flow constants are taken as known. The two edge probes are dropped, so the O1 pipeline then runs on the inner 18 probes.

Known-answer tests. On a fresh solver run (default physics, $k = 0.002\,\mathrm{s}^{-1}$, $N_z = 200$, horizon 3 t_{stoich}, 300 times), the inversion recovers q to within 3 % of $\max q$ over the inner cells. Fed the true q, the O2 pipeline recovers L1 with $|\hat{k} - k|/k < 0.02$ (weak form, M4). With the inverted q it does not recover the law at this Da, which the pre-registration declared as a pilot fact.

M5: errors-in-variables best-subset (an oracle)

M5 runs M4's enumeration of 1 to 4 terms on Lib-B in the strong form, but fits each support by mixed least squares and total least squares. The constant column is treated as exact and projected out by least squares. The other columns and the target are scaled by their error standard deviations and solved by total least squares through the SVD. The selection uses the same floored BIC as M4. The error standard deviations are the declared ones: σ times the observed channel range for c, q and T; for q^*_{meas}, $\sqrt{(\varepsilon\, q^*_{\mathrm{meas}})^2 + (\partial q^*_{\mathrm{meas}}/\partial c\ \sigma_c)^2}$, with the derivative taken by a forward difference of step 10^{-6} times the range of c; first-order propagation for cq, q^2 and c^2; and for the target, the q-channel standard deviation times the norm of the Savitzky–Golay derivative coefficients (window 21, order 3). At $\sigma = 0$ the target standard deviation is set to 10^{-12}. Because M5 is given the true error levels, it takes no part in choosing the best method for H2a, H2b or the positive control, and is used in H2c only.

Known-answer tests. With $y = 2x + 1$ and noise of standard deviation 0.15 on x (20 000 points), ordinary least squares is attenuated by more than 0.2 in the slope and mixed least squares and total least squares recovers slope and intercept within 0.05. At near-zero declared error (10^{-6} of each column's standard deviation) on the known-answer system, M5 recovers L1 and selects the same support as M4. Through the grid wrapper at zero noise on the five-probe system, M5 recovers L1 and agrees with M4.

ODR-BINDy and WENDy were not run on the grid. The ODR-BINDy port recovered the exact 7-term support on its authors' Lorenz example in 1 of 5 noise seeds, with a mean relative coefficient

error of 0.166 against about 2.2×10^{-3} reported, and is reported as "port, accuracy below the reference".

M6: SINDy-PI on the isothermal control

M6 is given no isotherm and no temperature. It runs only on the 9 isothermal-control cells, at every σ and at $\varepsilon = 0$, with 60 replicates, in the strong form. The library is $\{1, c, q, cq, q^2, c^2\}$, augmented with $\mathrm{d}q/\mathrm{d}t$, $c\,\mathrm{d}q/\mathrm{d}t$ and $q\,\mathrm{d}q/\mathrm{d}t$. Each of the three $\mathrm{d}q$ terms is tried as the left-hand side and fitted on the rest by M4's enumeration with up to 5 terms. The candidate with the smallest relative residual wins, and the relation is normalised so the $\mathrm{d}q/\mathrm{d}t$ coefficient is 1. Success requires the support $\{\mathrm{d}q, c\,\mathrm{d}q, c, q, cq\}$ exactly, with $c\,\mathrm{d}q > 0$, $c < 0$, $q > 0$ and $cq > 0$.

Known-answer test. On the known-answer system, M6 returns exactly the support above, and every coefficient is within 5 % of its true value ($c\,\mathrm{d}q = 3$, $c = -0.3$, $q = 0.02$, $cq = 0.06$). The errors measured on 2026-10-04 were 0.02 % and 0.07 % for b and k. Through the grid wrapper on the five-probe system, M6 succeeds.

Grid result. On the isothermal control M6 succeeded in none of its 2,160 fits (9 Da levels, 4 noise levels, 60 replicates), at every noise level including none. Every fit kept six terms, the five of the implicit relation's size plus one, so the exact support was never returned; the selected terms changed with noise and Da. We did not diagnose why the selection floor that passes the five-probe known-answer test admits a sixth term on the column data.

M7: symbolic regression (PySR)

M7 runs PySR with binary operators $+$, $-$ and $\times$, no unary operators, 40 iterations, maximum expression size 12, a fixed random state equal to the replicate index, deterministic serial search, PySR's "best" model selection and a time limit of 1800 s per search. The features are Lib-B without the constant, since PySR fits constants itself, and the target is the strong-form derivative. Feature names are passed as placeholders $f_0, f_1, \ldots$ and mapped back. The chosen expression is expanded by SymPy. A monomial that is a single feature to the first power counts under that feature; any other monomial is recorded as extra support, so a spurious term makes the fit fail. M7 runs on the subgrid of all Da, $\sigma \in \{0.5, 2\}$ %, $\varepsilon \in \{0, 2\}$ %, both isotherms, Pe $\times$ 1, O1, with 60 replicates.

Inclusion gate. Before any M7 search on the grid, PySR with these settings ran on the known-answer system with 0.5 % relative noise on q and on q^*_{meas}, for 3 seeds. The rule was to include M7 if L1 was recovered in at least 2 of 3 seeds, every search finished within 1800 s and peak memory stayed under 3 GB. All three seeds returned $k(q^*_{\mathrm{meas}} - q)$ with $\hat{k} = 0.019994$, 0.019992 and 0.019943 (relative errors 3.1×10^{-4}, 4.2×10^{-4} and 2.9×10^{-3}), in 35, 5 and 5 s, at a peak of 0.97 GB. M7 was included.

M8: Kolmogorov–Arnold network (not run on the grid)

M8 uses pykan 0.2.8 following Liu et al.'s procedure. It fits a KAN of width $[n, 1]$ (grid 5, spline order 3) on standardised raw variables $\{c, q, T, q^*_{\mathrm{meas}}\}$ with L-BFGS for 100 steps and sparsity penalty 10^{-3}, snaps every edge to $\{x, x^2, 0\}$, and reads the formula back in the original variables. It is not a reimplementation of KANDy.

Known-answer test and cost gate. M8 failed its known-answer test at zero noise in every configuration tried: a polynomial library at three sparsity penalties (10^{-3}, 10^{-2}, 10^{-1}), and raw variables on one and on three trajectories. It routed the law through q^2, used q alone, or returned $\{1, q^*_{\mathrm{meas}}, (q^*_{\mathrm{meas}})^2\}$. The test is kept as an expected failure. Its grid run was gated on one fit at the grid's size (20 probes, 200 times) taking at most 300 s. The measured fit took 1144 s, which projects to 68.7 h for the 216 subgrid fits, and returned only a constant. M8 was therefore not run on the grid.

M9: slow-manifold discovery

M9 regresses the state q, not its rate, on the q-free features $\{1, c, T, c^2, q^*_{\mathrm{meas}}\}$ by M4's enumeration with up to 3 terms and the same floored BIC. It runs on every non-isothermal L1 cell, O1, every (σ, ε), 60 replicates. Success is the support $\{q^*_{\mathrm{meas}}\}$ with coefficient 1 within 5 %.

Known-answer tests. On the known-answer system, M9 recovers $q = q^*_{\mathrm{meas}}$ at $k = 100\,\mathrm{s}^{-1}$ and correctly does not at $k = 0.002\,\mathrm{s}^{-1}$. Through the grid wrapper on the five-probe system at $k = 20\,\mathrm{s}^{-1}$, it succeeds.

M1: universal differential equation (demonstration)

M1 places a network in the uptake term of the column model. The network has two hidden layers of 32 tanh units. Its inputs are c/c_{in}, $q/q_{\max}$, $(T - T_{\mathrm{in}})/10\,\mathrm{K}$ and $q^*_{\mathrm{meas}}/q_{\max}$, and its output is multiplied by $q_{\max}/t_{\mathrm{stoich}}$, which needs no knowledge of k. Weights start as 0.3 times standard normal draws from a seed equal to the replicate index, and biases start at zero. The network is trained through a differentiable column ($N_z = 50$, Kvaerno5, PID step control with rtol 10^{-6} and atol 10^{-9}, initial step 10^{-3} s, at most 200 000 steps). The loss is the mean squared error of the outlet $c(t)$ and $T(t)$, each divided by its observed range. The optimiser is SciPy's L-BFGS-B with at most 50 iterations, 60 function evaluations and 7200 s, keeping the best parameters seen. If a column solve exhausts its step budget after the first evaluation, training stops and keeps the best parameters (correction B88). The trained rate over all solved states is read out by M4 on Lib-B and by PySR with M7's settings; success on either counts. M1 ran once: Langmuir, Pe × 1, $\sigma = 2$ %, $\varepsilon = 0$, O3, the 9 Da levels, one replicate.

Pilot and tests. The JAX column matches the reference solver to better than 5×10^{-3} on the exit curve, and $N_z = 50$ differs from $N_z = 400$ by 0.55 %. One gradient with respect to a 2×32 network took 145 s, about 50 steps in 2 h, so M1 could not run at the declared scale and was reduced to the demonstration above. A smoke test on a fresh column ($k = 0.002\,\mathrm{s}^{-1}$) runs two L-BFGS steps and checks that the loss is finite and the readout returns; it does not test recovery. Three tests check the B88 rule: training keeps the best parameters when the solver fails, and a failure on the first evaluation is still an error. M1 has no known-answer recovery test.

Success metrics

L1 succeeds when the selected support is exactly $\{q^*_{\mathrm{meas}}, q\}$, the q^*_{meas} coefficient is positive, the q coefficient is negative, and their ratio is -1 within 10 %. The fitted rate constant is $\hat{k} = -\mathrm{coef}(q)$. For L2, with q^*_{meas} fixed, the implied rate $\hat{k}(q) = -(\mathrm{coef}(q) + \mathrm{coef}(cq)\,c + 2\,\mathrm{coef}(q^2)\,q)$ is averaged in 10 equal bins of $q/q_{\max}$; the state dependence is found when $\{q^*_{\mathrm{meas}}, q\}$ are selected, the Spearman correlation over the bins is below -0.8, and the top bin is at most 0.6 of the bottom bin. A fit that raised an error counts as a failure.

Profile-likelihood identifiability

Identifiability is judged with the true law known and with the same measured isotherm the methods receive. k is practically identifiable when the 95 % profile interval, the set where $2(\max \ell - \ell) \leq 3.84$, lies within $[k/2, 2k]$. All profiles refine the same way: after the coarse grid, a pass of log-spaced points (41 on O1, 21 on O2 and O3) is placed between the outside neighbours of the current interval, for at most 6 passes, stopping when at least 5 points lie inside or the step ratio falls below 1.001. The reported ends are the outside neighbours, so the interval is conservative. As first run, the O1 profile used 21 points per pass; it was corrected to the pre-registered 41 before the corrected O1 rows were computed (Supplementary Note 2).

O1. The log-likelihood is the Gaussian sum over all 20 probes. For each probe, $q^*(t)$ is built by PCHIP interpolation of the observed c and T on a grid 4 times finer than the observations, and the LDF law is integrated by its exact update for a piecewise-linear q^*, vectorised over k. The coarse grid is $k =$ `logspace(-7, 1, 81)`, 10 points per decade, more than a decade beyond the smallest and largest k of the design. If the coarse likelihood peaks at either end of that grid, the profile raises an error, which counts as not identifiable. The noise standard deviation in the likelihood is $\max(\sigma, 10^{-4})$ times the range of the observed q. O1 identifiability runs on every non-isothermal L1 cell, every (σ, ε), 60 replicates.

O2 and O3. Each point of the profile is a full column solve at $N_z = 100$ with the measured isotherm, scored on the interior c and T probes (O2) or the outlet c and T (O3). The coarse grid is $k =$ `logspace(-6, 0, 25)`. The noise standard deviations are $\max(\sigma, 10^{-4})$ times the ground-truth range of c and of T. Each solve is capped at 600 s of wall time; a capped solve scores $-\infty$, and each row records the capped k values and whether one lies within one coarse step ($10^{6/24}$) of the interval. These run on the subgrid of all Da, both isotherms, Pe $\times$ 1, L1, $\sigma \in \{0.5, 2\}$ %, $\varepsilon \in \{0, 2\}$ %, with 3 replicates (216 rows per observation model).

Known-answer tests. The exact O1 update agrees with `solve_ivp` on a piecewise-linear q^* for kh from 10^{-4} to 100 (a test added with the B90 correction; the earlier test, against the ODE-solver likelihood at $k \in \{0.005, 0.02, 0.05\}$, had too small a kh to detect the error). The O1 profile over 20 noisy probes ($\sigma = 2$ %) contains $k = 0.02$ and is identifiable. After A6, it contains k at 3.22×10^{-6}, 1.39×10^{-5} and $0.139\,\mathrm{s}^{-1}$, and it raises an error on a grid whose edge holds the peak. The single-probe profile on a fresh non-isothermal run ($k = 2 \times 10^{-4}\,\mathrm{s}^{-1}$, $\sigma = 0.5$ %) lies within 2 % of k. The outlet and probe profiles on a fresh low-Da run ($k = 2 \times 10^{-4}\,\mathrm{s}^{-1}$) contain k with hi/lo < 2. The solve cap scores a capped solve as $-\infty$, records it, and leaves an uncapped solve unchanged.